

The Boltzmann equation for conduction electrons

By P. L. TAYLOR†

Cavendish Laboratory, University of Cambridge

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Methods of solution of the Boltzmann equation for the distribution of electrons in an anisotropic metal are discussed. Some general properties of integral equations are used to develop a method suitable for use in numerical calculations.

1. INTRODUCTION

In the theory of electrical conduction in metals it is common to assume the existence of a relaxation time, τ ; one supposes that in the absence of any applied fields an electric current decays as $\exp(-t/\tau)$. This is only an approximation to the behaviour of a real metal. A real metal has a crystal lattice structure, and is consequently anisotropic. The decaying electric current may then change direction as well as magnitude, and such behaviour cannot be described by a single constant, τ . Instead, a transport equation must be set up and solved. The purpose of the present paper is to discuss the solution of a simple form of such a transport equation in an anisotropic metal, and to suggest a practical method whereby the problem may be solved most conveniently.

We take as our starting point the *linearized Boltzmann equation*, which defines the probability, $f_{\mathbf{k}}$, that the electron state of crystal momentum $\mathbf{k}$ is occupied. We consider a magnetic field, $\mathbf{H}$, and an electric field $\mathbf{E}$ applied to a specimen in which there is a temperature gradient ∇T . The energy of an electron in the state $\mathbf{k}$ is $\mathcal{E}_{\mathbf{k}}$ and the velocity $\mathbf{v}_{\mathbf{k}} = \hbar^{-1} \partial \mathcal{E} / \partial \mathbf{k}$. The occupation function in equilibrium, $f_{\mathbf{k}}^0$ is given in terms of the chemical potential, ζ , and the thermal energy, kT , by the usual Fermi function. The Boltzmann equation is then (Ziman 1960, chap. VII)

$$\mathbf{v} \frac{\partial f^0}{\partial \mathcal{E}} \cdot (e\mathbf{E} - \nabla \zeta) - (\mathcal{E} - \zeta) \frac{\nabla T}{T} \frac{\partial f^0}{\partial \mathcal{E}} \cdot \mathbf{v} + \frac{e}{\hbar c} \frac{\partial}{\partial \mathbf{k}} \cdot f_{\mathbf{k}} (\mathbf{v} \times \mathbf{H}) = \frac{df_{\mathbf{k}}}{dt}. \quad (1)$$

If in equilibrium the probability of an electron being scattered from $\mathbf{k}$ to $\mathbf{k}'$ is $P(\mathbf{k}, \mathbf{k}')$, then the right-hand side of equation (1) is (Wilson 1953, §8.1)

$$\frac{df}{dt} = \int P(\mathbf{k}, \mathbf{k}') \left[\frac{f_{\mathbf{k}'}^1}{f_{\mathbf{k}'}^0 (1 - f_{\mathbf{k}'}^0)} - \frac{f_{\mathbf{k}}^1}{f_{\mathbf{k}}^0 (1 - f_{\mathbf{k}}^0)} \right] d\mathbf{k}', \quad (2)$$

where summation over states is represented by $\int d\mathbf{k}$ and $f_{\mathbf{k}}^1 = f_{\mathbf{k}} - f_{\mathbf{k}}^0$. To illustrate the method of solution of this equation we shall first consider the simplest non-trivial case, in which an electric field alone is applied to a metal in which the scattering of electrons is completely elastic. In this case equations (1) and (2) reduce to

$$\mathbf{v} \cdot e\mathbf{E} \frac{\partial f^0}{\partial \mathcal{E}} = \int Q(\mathbf{k}, \mathbf{k}') (f_{\mathbf{k}'} - f_{\mathbf{k}}) d\mathbf{k}', \quad (3)$$

† Present address: Department of Physics, Case Institute of Technology, Cleveland 6, Ohio, U.S.A.

where $Q(\mathbf{k}, \mathbf{k}')$ is the scattering probability when the state $\mathbf{k}$ is known to be occupied and $\mathbf{k}'$ to be unoccupied. A further simplification is achieved by defining a *vector mean free path*, $\Lambda(\mathbf{k})$, by the equation

$$e\mathbf{E} \cdot \Lambda(\mathbf{k}) \partial f^0 / \partial \mathcal{E} = -f_{\mathbf{k}}^1.$$

Substitution of this into (4) shows that Λ may be taken to be independent of $\mathbf{E}$, and provides a complete description of the conductivity of the metal: this description of the electron distribution function in the presence of the electric field is the most concise way possible, there being just three independent numbers for each point in k -space. It is more convenient than the introduction of a tensor relaxation time (Dumke 1956; Herring & Vogt 1956), which contains nine components of which only three can be independent. The electric current is given by

$$\begin{aligned} \mathbf{j} &= \int e f_{\mathbf{k}}^1 \mathbf{v}_{\mathbf{k}} d\mathbf{k} \\ &= -e^2 \int \mathbf{v}_{\mathbf{k}} (\Lambda \cdot \mathbf{E}) \frac{\partial f^0}{\partial \mathcal{E}} d\mathbf{k}. \end{aligned}$$

The conductivity tensor, σ , is then

$$\sigma = -e^2 \int \mathbf{v} \Lambda \frac{\partial f^0}{\partial \mathcal{E}} d\mathbf{k}.$$

For metals having cubic symmetry this reduces to a scalar

$$\sigma = \frac{1}{3} \text{tr } \sigma = -\frac{1}{3} e^2 \int \mathbf{v} \cdot \Lambda \frac{\partial f^0}{\partial \mathcal{E}} d\mathbf{k}. \quad (4)$$

In terms of $\Lambda(\mathbf{k})$, the Boltzmann equation, (3), is now

$$\mathbf{v} = \int Q(\mathbf{k}, \mathbf{k}') (\Lambda_{\mathbf{k}} - \Lambda_{\mathbf{k}'}) d\mathbf{k}'. \quad (5)$$

Our problem is the solution of this for Λ in terms of $\mathbf{v}$ and $Q(\mathbf{k}, \mathbf{k}')$.

2. THE SOLUTION OF THE BOLTZMANN EQUATION

We abbreviate our notation by defining the linear operator, Q ,

$$Q x(\mathbf{k}) \equiv \int Q(\mathbf{k}, \mathbf{k}') x(\mathbf{k}') d\mathbf{k}'$$

and the function $q(\mathbf{k})$,

$$q(\mathbf{k}) \equiv Q \cdot 1 \equiv \int Q(\mathbf{k}, \mathbf{k}') d\mathbf{k}'.$$

In terms of these (5) becomes

$$\mathbf{v} = (q - Q) \Lambda. \quad (6)$$

This may be rearranged to give

$$\Lambda = q^{-1} Q \Lambda + q^{-1} \mathbf{v}. \quad (7)$$

It is now clear that the Boltzmann equation is an example of an inhomogeneous Fredholm equation of the second kind, having a positive definite polar kernel. It is a special case of the eigenvalue equation

$$\Lambda = \lambda q^{-1} Q \Lambda + q^{-1} \mathbf{v}.$$

The existence of a solution of this for $\lambda = 1$ is assured by the symmetry properties of Q and $\mathbf{v}$.

For a general function $\mathbf{v}(\mathbf{k})$ subject to the solubility condition, an equation such as (7) cannot in general be solved in closed form. The solution is expressible in terms of the *Green function*, or *solving kernel* of the integral equation. In order to find the Green function, however, it is necessary to know the solutions of the homogeneous equation

$$\Lambda q = \lambda Q \Lambda \quad (8)$$

and the corresponding eigenvalues λ . This is only possible by expanding $Q(\mathbf{k}, \mathbf{k}')$ in a set of known functions and solving the (generally infinite) set of linear algebraic equations obtained for the coefficients in the expansion (Morse & Feshbach 1953, chap. VIII).

Of the methods that have previously been used in the solution of the Boltzmann equation, the simplest is the relaxation time approximation. Here it is assumed that the operator $q - Q$ may be replaced by a constant. Equation (6) then becomes

$$\mathbf{v} = \tau^{-1} \Lambda,$$

where the constant τ is known as the relaxation time. Because the conductivity of a cubic metal, as given by equation (4), depends only on the scalar product of the electron velocity and the mean free path, it is always possible to define a function $\tau(\mathbf{k})$ by

$$\mathbf{v} \cdot \Lambda = \tau v^2$$

which will give the correct value of the conductivity when the approximation

$$\Lambda = \tau \mathbf{v}$$

is used. Then $\tau(\mathbf{k})$ is known as an *anisotropic relaxation time*. Intuitive formulae from which it might be calculated have been given by Ziman (1961) and Deutsch, Paul & Brooks (1961). Variational and Green function procedures have also been suggested and used (Kohler 1948; Sondheimer 1950, 1962; Price 1957) for the evaluation of $\tau(\mathbf{k})$.

Unfortunately, none of these methods is suitable for the investigation of the transport properties of metals with highly anisotropic Fermi surfaces. Ziman's formula for the anisotropic relaxation time is based only on intuitive arguments, and has indeed been shown by Sondheimer (1962) to give results incorrect both in sign and order of magnitude for the variation of $\tau(\mathbf{k})$ in one exactly soluble model. Use of Sondheimer's method is not always convenient because it is not possible to analyze the scattering probability in the way required by that method without taking a very large number of terms to account for crystal symmetry. When the anisotropy of the system is small the variational method is appropriate, although it is always difficult to estimate the accuracy of the results so obtained. When the anisotropy is large this method becomes very cumbersome, especially when the energy surfaces are not described by simple analytic functions. It is thus necessary to find some other method of solution.

3. THE SCATTERING EIGENFUNCTIONS

The method which will be described here for solving the Boltzmann equation starts by obtaining the formal iterative solution of (7). This is

$$\begin{aligned}\Lambda &= q^{-1}\mathbf{v} + q^{-1}Q q^{-1}\mathbf{v} + \dots \\ &= \sum_{n=0}^{\infty} (q^{-1}Q)^n q^{-1}\mathbf{v}.\end{aligned}\quad (9)$$

Because $(q - Q)$ is singular and non-diagonal, the series always converges; the formal proof of convergence is given in the Appendix. Although the series (9) is always convergent, it is the rapidity with which it does so on which its convenience as a means of numerical solution depends. In the trivial case of an isotropic metal the operator $q^{-1}Q$ is just a multiplying factor of $\overline{\cos \theta}$, the average cosine of the angle through which an electron is scattered, and we expect the convergence to be similar in the anisotropic case. Thus when the scattering itself is isotropic—that is, when $Q(\mathbf{k}, \mathbf{k}')$ is just a constant—the convergence of the series is immediate, but when the scattering is confined to states $\mathbf{k}'$ very near to $\mathbf{k}$ on the Fermi surface—as would be the case for scattering by lattice vibrations at low temperatures—then the series may converge very slowly, and be quite unsuited to direct numerical computation. However, it will now be shown that enough information can generally be extracted from the first few terms of (9) to give $\Lambda(\mathbf{k})$ accurately for all forms of $Q(\mathbf{k}, \mathbf{k}')$.

The operators Q and $q^{-1}Q$ are both positive definite because $Q(\mathbf{k}, \mathbf{k}')$, being a probability, is always positive. We define the complete set of eigenvalues and eigenfunctions of $q^{-1}Q$ by

$$q^{-1}Q \mathbf{a}_r(\mathbf{k}) = \alpha_r \mathbf{a}_r(\mathbf{k}).$$

We refer to the $\mathbf{a}_r(\mathbf{k})$ as the *scattering eigenfunctions*. It is shown in the Appendix that α_0 , the greatest of the eigenvalues α_r , has the value unity and corresponds to the eigenfunction $\mathbf{a}_0(\mathbf{k}) = \text{constant}$. It follows from the fact that $q^{-1}Q$ is positive definite that all the α are positive; also, since $\mathbf{k}$ is confined to a surface of constant energy in k -space we know that the larger α correspond to the eigenfunctions $\mathbf{a}_r(\mathbf{k})$ which have least nodal lines on this surface.

We call $q^{-1}\mathbf{v}(\mathbf{k}) \equiv \mathbf{y}(\mathbf{k})$, and expand this in terms of the $\mathbf{a}_r(\mathbf{k})$. We normalize the $\mathbf{a}_r(\mathbf{k})$ by the condition

$$\int q(\mathbf{k}) [\mathbf{a}_r(\mathbf{k})]^2 \delta(\mathcal{E} - \mathcal{E}_f) d\mathbf{k} = 1 \quad (10)$$

and write

$$\mathbf{y}(\mathbf{k}) = \sum_r \beta_r \mathbf{a}_r(\mathbf{k}).$$

Then

$$(q^{-1}Q)^n \mathbf{y}(\mathbf{k}) = \sum_r \beta_r \alpha_r^n \mathbf{a}_r(\mathbf{k}). \quad (11)$$

We call this quantity $\mathbf{y}_n(\mathbf{k})$. The solution of the Boltzmann equation is then given by

$$\begin{aligned}\Lambda &= \sum_n \mathbf{y}_n(\mathbf{k}) \\ &= \sum_{n,r} \beta_r \alpha_r^n \mathbf{a}_r(\mathbf{k}) \\ &= \sum_{r=1}^{\infty} \frac{\beta_r}{1 - \alpha_r} \mathbf{a}_r(\mathbf{k}).\end{aligned}\quad (12)$$

The coefficient of the term in $\mathbf{a}_0(\mathbf{k})$ in the expansion is not determined because $\mathbf{a}_0(\mathbf{k})$ is a solution of the homogeneous equation (8). However, this term makes no contribution to the current and may thus be neglected.

Were $q^{-1}Q$ a symmetric operator, the eigenfunctions $\mathbf{a}_r(\mathbf{k})$ would be orthogonal—that is, we should have

$$\int \mathbf{a}_r(\mathbf{k}) \cdot \mathbf{a}_s(\mathbf{k}) \delta(\mathcal{E} - \mathcal{E}_f) d\mathbf{k} = 0 \quad (r \neq s). \quad (13)$$

However, although Q is symmetric, q^{-1} is not. (I am indebted to Professor E. H. Sondheimer for pointing this out.) A set of functions which are orthogonal can be formed by multiplying the $\mathbf{a}_r(\mathbf{k})$ by $[q(\mathbf{k})]^{1/2}$. Then it can be shown that

$$\int \mathbf{a}_r(\mathbf{k}) \cdot \mathbf{a}_s(\mathbf{k}) q(\mathbf{k}) \delta(\mathcal{E} - \mathcal{E}_f) d\mathbf{k} = 0 \quad (r \neq s). \quad (14)$$

With the normalization condition (10) the integral is equal to unity when $r = s$. Let us denote the integral

$$\int x(\mathbf{k}) y(\mathbf{k}) q(\mathbf{k}) \delta(\mathcal{E} - \mathcal{E}_f) d\mathbf{k}$$

for any pair of real functions $x(\mathbf{k})$, $y(\mathbf{k})$ by the symbol

$$(x|y).$$

Then from (11)

$$(y_m|y_n) = \sum_r \beta_r^2 \alpha_r^{m+n}. \quad (15)$$

We now have the basis of a method of finding Λ , if we can assume that y can be expanded in just a few of the scattering eigenfunctions. If $y_0, y_1, \dots, y_m$ are calculated, then $2m+1$ equations of the form (15) can be constructed, with which an expansion of y_0 into m eigenfunctions can be solved completely. In fact, $2m$ equations would serve to determine the coefficients uniquely, although the proof that this is so is not trivial, and rests on the fact that all the terms in equations (15) are positive. The extra equation which overdetermines the coefficients may then be considered as an aid to the numerical solution. In practice, however, the great advantage of the iteration method is found to lie in the fact that not only do the β decrease rapidly with increasing r , but so also do the α . The necessity of solving equations (15) explicitly, which would normally be done by successive approximation, may then be avoided. It suffices to continue the iteration until, to any desired accuracy

$$(y_{m-1}|y_{m-1})(y_m|y_m) - (y_m|y_{m-1})^2 \simeq 0. \quad (16)$$

This quantity is, from equation (15), equal to

$$\sum_{p>r} [\beta_r \beta_p (\alpha_p - \alpha_r) \alpha_p^{m-1} \alpha_p^{m-1}]^2,$$

which, being a sum of squares, is always positive. When the sum is negligibly small it may then be assumed that each term in the summation is similarly small. We may then write, with an accuracy related to that with which equation (16) is obeyed, that

$$y_m = \alpha_1^m \beta_1 a_1(\mathbf{k}).$$

We now split the iteration series for Λ into two parts.

$$\Lambda = \sum_{n=0}^{m-1} y_n(\mathbf{k}) + \sum_{n=m}^{\infty} y_n(\mathbf{k}).$$

The second part we now know to be a geometric series, since y_m contains only one eigenfunction, and so we can write this as $y_m(k)/(1-\alpha_1)$. The coefficient α_1 , is then simply given by $(y_m|y_m)/(y_m|y_{m-1})$. Thus m iterations have sufficed to give us an expression for the mean free path,

$$\Lambda = \sum_{r=0}^{m-1} y_r + \frac{(y_m|y_{m-1})}{(y_m|y_{m-1}) - (y_m|y_m)} y_m.$$

The accuracy of this expression is, at the very least, greater than that of the relation (16).

The only real test of the value of the method must be an examination of the results it produces for any particular case. We cannot intuitively estimate the number of terms required in the expansion of y_0 , for instance, or the way in which the β decrease with increasing r . We can, however, make some general comments on the series of eigenvalues α_r , and argue that it is plausible that they should decrease rapidly with r . For isotropic scattering all the α are zero, while when $Q(\mathbf{k}, \mathbf{k}')$ approximates to a δ -function we have the least favourable case in which all the α tend to unity. If we examine the latter case, and suppose the scattering to be confined to a fine lobe around $\mathbf{k}' = \mathbf{k}$, we find that $1 - \alpha_r$ is proportional to the second derivative of $\mathbf{a}_r$ with respect to $\mathbf{k}$ over the surface. The problem is similar to that of finding the wave functions of a particle in a box, and consequently $1 - \alpha_r$ is proportional to r^2 . Thus even in this, the least favourable of examples, the convergence of (12) is rapid.

The extension of this method to the case of inelastic scattering is trivial; a modification of the orthogonality conditions is required, however. Because surfaces of constant energy are then no longer independent the scattering eigenfunctions must be defined at all points in k -space, and the method becomes unsuitable for use in numerical calculations.

5. INCLUSION OF A TEMPERATURE GRADIENT

When the scattering is elastic, this method of solution may be immediately extended to include the effect of a temperature gradient. It suffices to redefine Λ by

$$\left[e\mathbf{E} - (\mathcal{E} - \zeta) \frac{\nabla T}{T} \right] \cdot \Lambda \frac{\partial f^0}{\partial \mathcal{E}} = -f_k^1. \quad (17)$$

On substitution in (1) the term in brackets cancels and the simpler form of the Boltzmann equation (6) is regained. Thus when there is only elastic scattering, as is predominantly the case in impure metals at low temperatures (Taylor 1962), a unique mean free path exists which describes the effect of both electric fields and temperature gradients. The Wiedemann-Franz law is consequently obeyed under these conditions.

For inelastic scattering the solution is more complicated. In this case we must return to equation (2) and define an operator P and a function $p(\mathbf{k})$ analogous to Q and q .

$$Px(\mathbf{k}) \equiv \int P(\mathbf{k}, \mathbf{k}') x(\mathbf{k}') d\mathbf{k}'; \quad p(\mathbf{k}) \equiv P.1.$$

In terms of these the Boltzmann equation for an electric field alone becomes

$$\mathbf{v}f_0^0(1-f_0^0) = (p-P)\Lambda. \tag{18}$$

This form of the equation cannot be obtained when there is a temperature gradient, because P does not commute with the electron energy, $\mathcal{E}$. For instance, when a large electric field and a small temperature gradient in the same direction are simultaneously applied, the Boltzmann equation in terms of the Λ defined in (17) is

$$\mathbf{v}f_0^0(1-f_0^0) \simeq (p-P)\Lambda + \int P(\mathbf{k}, \mathbf{k}') \Lambda_{\mathbf{k}'} \left(\frac{\mathcal{E}' - \mathcal{E}}{eET} \right) \nabla T d\mathbf{k}'.$$

This differs from (18) in the presence of an interference term on the right-hand side of the equation. The transport equation must consequently be solved for every combination of applied electric field and temperature gradient individually.

6. SOLUTION IN THE PRESENCE OF A MAGNETIC FIELD

When an electric and a magnetic field are simultaneously applied the Boltzmann equation is of the form (1). The change in the electron distribution function is still proportional to the applied electric field, and so we write

$$f_{\mathbf{k}}^1 = -(\partial f_0^0 / \partial \mathcal{E}) e\mathbf{E} \cdot (\Lambda + \mathbf{M} + \mathbf{N} + \dots)$$

where Λ is independent of the magnetic field $\mathbf{H}$, and $\mathbf{M}$ is proportional to $\mathbf{H}$, $\mathbf{N}$ to $\mathbf{H}^2$, etc. This approach is a development of that employed by Jones & Zener (1934). Substituting this into (1) and equating powers of $\mathbf{H}$ give

$$\begin{aligned} \mathbf{v} &= (q-Q)\Lambda, \\ -(\hbar c)^{-1} [\mathbf{vH}\nabla_{\mathbf{k}}] \Lambda &= (q-Q)\mathbf{M}, \\ -(\hbar c)^{-1} [\mathbf{vH}\nabla_{\mathbf{k}}] \mathbf{M} &= (q-Q)\mathbf{N}, \\ &\dots\dots\dots \end{aligned}$$

Here $[\mathbf{vH}\nabla_{\mathbf{k}}]$ is the triple scalar product of the electron velocity, the magnetic field and the gradient operator in momentum space; it is a scalar operator which differentiates its operand around an orbit on the Fermi surface. The set of equations for the components of the mean free path may now be solved in turn. First, Λ is found as in the case where no magnetic field is applied; $\mathbf{M}$ may then be obtained from Λ by differentiating around an orbit and then operating upon this with the inverse of $(q-Q)$ as described in §3, and similarly for higher terms. The formal result is

$$\begin{aligned} f_{\mathbf{k}}^1 &= -\frac{\partial f_0^0}{\partial \mathcal{E}} e\mathbf{E} \cdot \sum_{n=0}^{\infty} \left\{ (q-Q)^{-1} \frac{-1}{\hbar c} [\mathbf{vH}\nabla_{\mathbf{k}}] \right\}^n (q-Q)^{-1} \mathbf{v} \\ &= -\frac{\partial f_0^0}{\partial \mathcal{E}} e\mathbf{E} \cdot \sum_{n=0}^{\infty} \left\{ \sum_{m=0}^{\infty} \left(\frac{Q}{q} \right)^m \frac{-1}{\hbar c q} [\mathbf{vH}\nabla_{\mathbf{k}}] \right\}^n \sum_{p=0}^{\infty} \left(\frac{Q}{q} \right)^p \frac{\mathbf{v}}{q}. \end{aligned}$$

Only terms up to H^2 need be considered in order to obtain the low-field limit of the galvanomagnetic tensor, and for cubic crystals there is enough cancellation to make this rather complicated expression manageable in cases of practical interest.

The high-field limit of the conductivity may be obtained by a similar approach in which $f_k^{(1)}$ is expanded in inverse powers of H . We write

$$f_k^1 = -(\partial f^0 / \partial \mathcal{E}) e \mathbf{E} \cdot (\mathbf{A} + \mathbf{B} + \mathbf{\Gamma} + \dots)$$

which, on substitution in (1), leads to

$$-(\hbar c)^{-1} [\mathbf{vH}\nabla_k] \mathbf{A} = \mathbf{0}, \quad (19a)$$

$$(q - Q) \mathbf{A} - (\hbar c)^{-1} [\mathbf{vH}\nabla_k] \mathbf{B} = \mathbf{v}, \quad (19b)$$

$$(q - Q) \mathbf{B} - (\hbar c)^{-1} [\mathbf{vH}\nabla_k] \mathbf{\Gamma} = \mathbf{0}, \quad (19c)$$

.....

The integration of the first expression is complicated by the fact that

$$[\mathbf{vH}\nabla_k] \mathbf{H} = \mathbf{0}$$

and so integration of (19a) gives the result

$$\mathbf{A} = \alpha \mathbf{H}$$

where α is any constant. However, since $\mathbf{A}$ is by definition independent of H , we must have $\mathbf{A} = \mathbf{0}$. Equation (19b) then tells us that $\mathbf{B}$ may be found by integrating the velocity around an orbit. Again $\mathbf{B}$ may contain a constant term in the H -direction; this may be neglected because a constant does not contribute to the current and does not affect the equation for $\mathbf{\Gamma}$. The series of equations (19) may thus be integrated directly term by term to give an expression for the perturbed distribution function in the presence of a strong magnetic field.

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APPENDIX

The mean free path may be written as the infinite sum

$$\Lambda = \sum_{n=0}^{\infty} \left(\frac{Q}{q} \right)^n y(\mathbf{k})$$

for each of the three Cartesian components of Λ and y . Now

$$\begin{aligned} \left| \frac{Q}{q} x(\mathbf{k}) \right| &= \left| \frac{\int Q(\mathbf{k}, \mathbf{k}') x(\mathbf{k}') d\mathbf{k}'}{\int Q(\mathbf{k}, \mathbf{k}') d\mathbf{k}'} \right| \\ &\leq \frac{\int Q(\mathbf{k}, \mathbf{k}') |x|_m d\mathbf{k}'}{\int Q(\mathbf{k}, \mathbf{k}') d\mathbf{k}'} = |x|_m, \end{aligned}$$

where $|x|_m$ is the greatest value taken by $|x|$ in the range of integration. The equality is only true when either

$$x(\mathbf{k}) = \text{constant}$$

or

$$P(\mathbf{k}, \mathbf{k}') = \delta(\mathbf{k} - \mathbf{k}').$$

Neither of these can apply when $x = (Q/q)^r y$ for any r , since the operator is an even function of $\mathbf{k}$ while y is an odd function of $\mathbf{k}$; and the condition $Q(\mathbf{k}, \mathbf{k}') = \delta(\mathbf{k} - \mathbf{k}')$ does not represent any scattering between states. Because Q operates on a closed surface the eigenvalues of Q/q are distinct, and

$$|(Q/q)x| \leq \alpha_1 |x|_m,$$

where α_1 is the second largest eigenvalue of Q/q . Consequently

$$\left| \sum_{n=0}^{\infty} \left(\frac{Q}{q} \right)^n y \right| \leq \frac{1}{1 - \alpha_1} |y|_m$$

and the series $\Sigma(Q/q)^n y$ always converges.

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